Chapter R Functions, Graphs and Models
MTH 201 - Business Calculus

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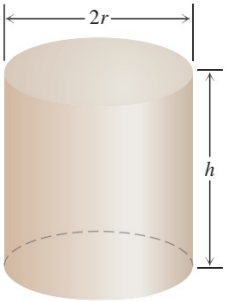
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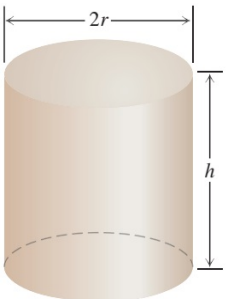
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Answer: $V = \pi r^2 h$, we find that the soup can has a radius of approximately 3.4 cm and a height of 6.8 cm.

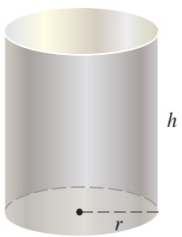
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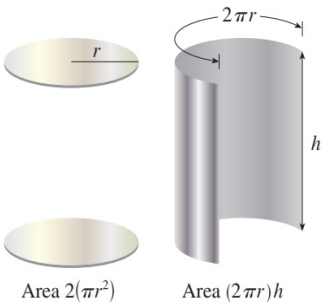
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Calculus problem: A soup can is to contain a volume of 250 cm^3. If the cost of the material for the two circular ends is \$0.0008 per square centimeter and the cost of the material for the side is \$0.0015 per square centimeter, what dimensions will minimize the cost of the can?

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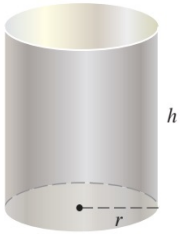


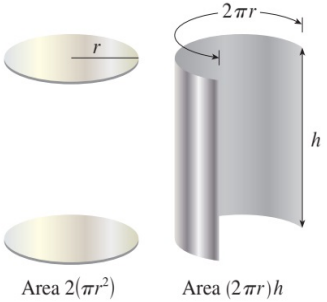


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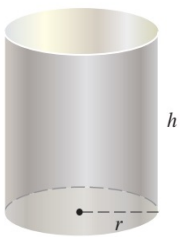


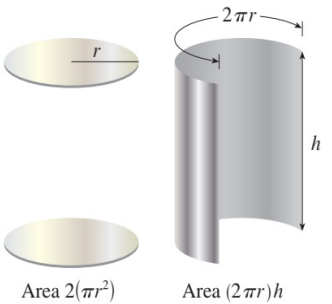


Answer: The total cost is $C = 0.0008 \times A_B + 0.0015 \times A_S$

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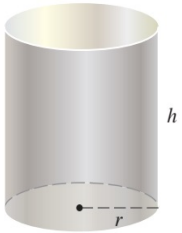


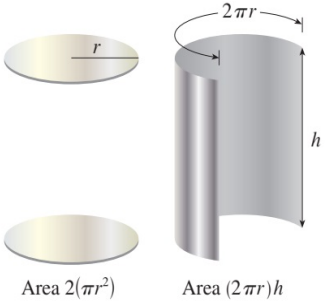
Answer: The total cost is $C = 0.0008 \times A_B + 0.0015 \times A_S = 0.0016\pi r^2 + \dfrac{0.75}{r}$

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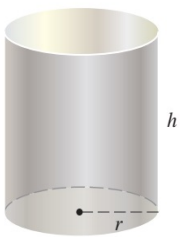


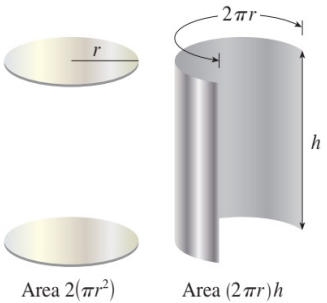


Answer: The total cost is $C = 0.0008 \times A_B + 0.0015 \times A_S = 0.0016\pi r^2 + \dfrac{0.75}{r} = C(r)$.

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Answer: The total cost is $C = 0.0008 \times A_B + 0.0015 \times A_S = 0.0016\pi r^2 + \dfrac{0.75}{r} = C(r)$.
Determine the value of r for which C has a minimum value. We need the tools of calculus to answer this.

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Outline

1. **Functions**
2. Graphs of functions
3. Nonlinear Functions and models
4. Mathematical modeling and curve fitting

Functions

Functions are the major objects we deal with in calculus because they are key to describing the real world in mathematical terms.

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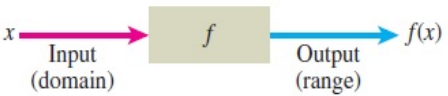
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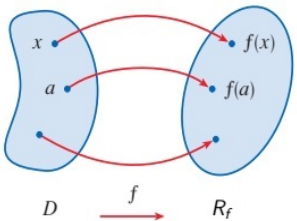
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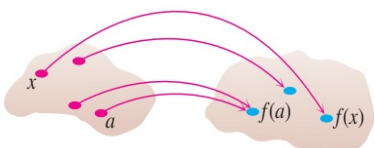
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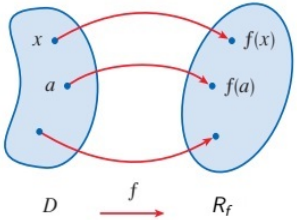


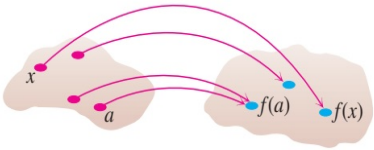


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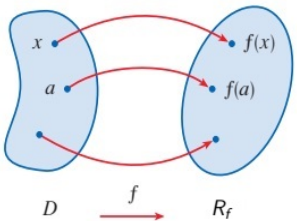


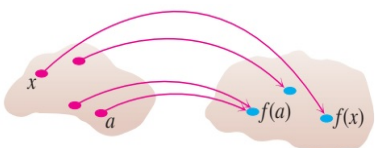


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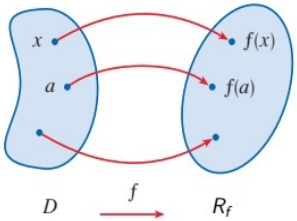


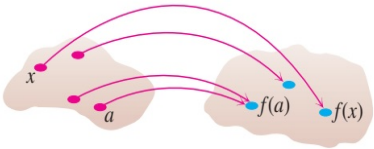
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- The set of all values of $f(x)$ as x varies throughout D is called the range of the function.

Functions

Exercise 1. Verify the domains and ranges of these functions:
a) $y = x^2$, b) $y = 1/x$, c) $y = \sqrt{x}$, d) $y = \sqrt{4 - x}$, e) $y = \sqrt{1 - x^2}$.

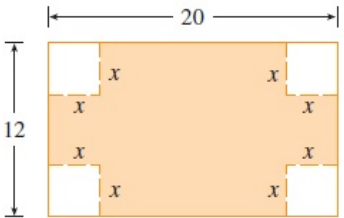
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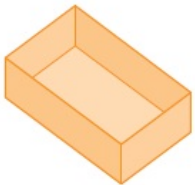
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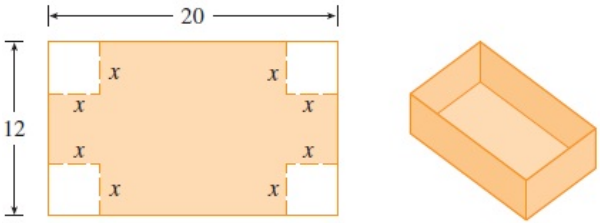


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Express the volume V of the box as a function of x.

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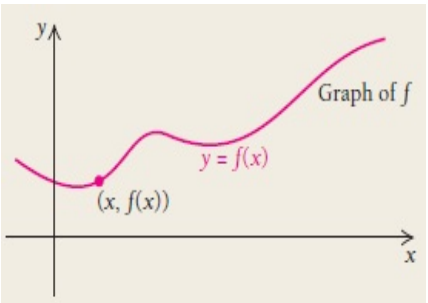
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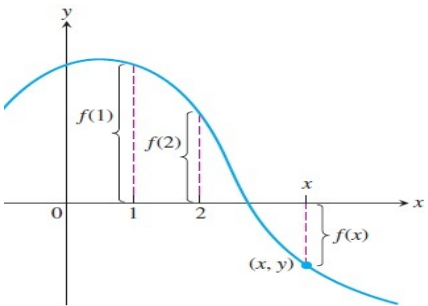
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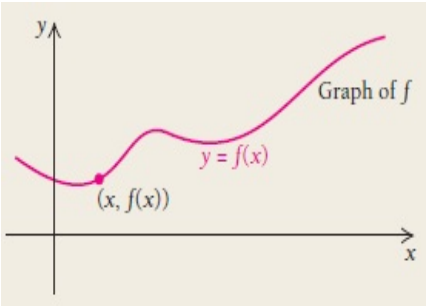


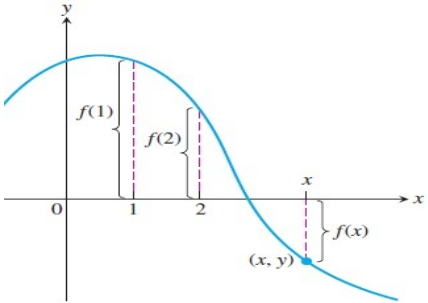
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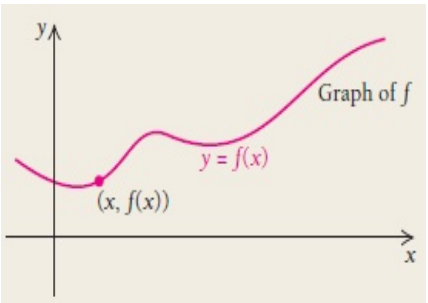


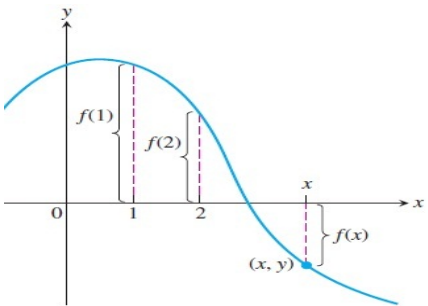
How to sketch the graph of a function?

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How to sketch the graph of a function? https://www.desmos.com

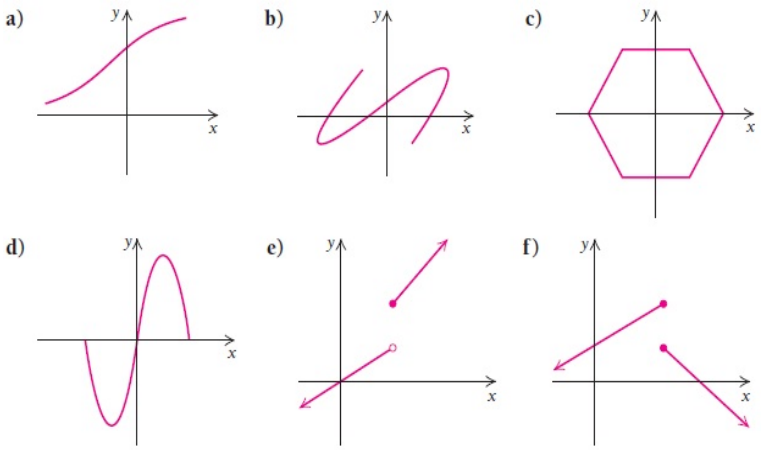
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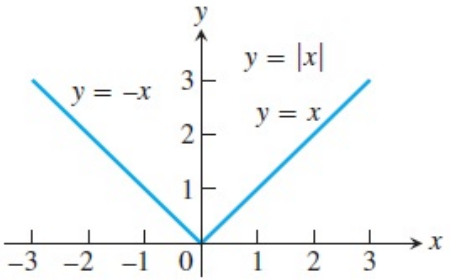
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Example. Graphing piecewise-defined function.

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The function

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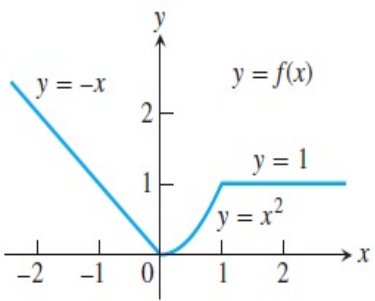
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Exercise 2. In a certain country, income tax is assessed as follows. There is no tax on income up to \$10000. Any income over \$10000 is taxed at a rate of 10%, up to an income of \$20000. Any income over \$20000 is taxed at 15%.

a) Sketch the graph of the tax rate R as a function of the income I.

b) How much tax is assessed on an income of \$14000? On \$26000?

c) Sketch the graph of the total assessed tax T as a function of the income I.

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Horizontal and Vertical Lines

The graph of $y = c$ or $f(x) = c$, a horizontal line, is the graph of a function. Such a function is referred to as a constant function. The graph of $x = a$ is a vertical line, and $x = a$ is not a function.

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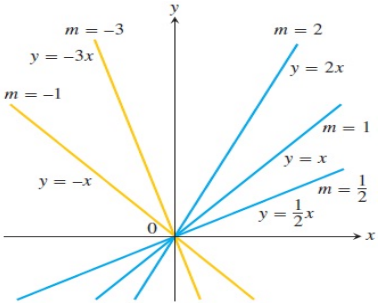
The graph of the function given by $y = mx$ or $f(x) = mx$ is the straight line through the origin $(0, 0)$ and the point $(1, m)$. The constant m is called the slope of the line.

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Horizontal and Vertical Lines

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The graph of the function given by $y = mx$ or $f(x) = mx$ is the straight line through the origin $(0, 0)$ and the point $(1, m)$. The constant m is called the slope of the line.



Slope and Linear Functions

Definition

Two variables y and x are **proportional** *(to one another) if one is always a constant multiple of the other, that is, if $y = mx$ for some nonzero constant m.*

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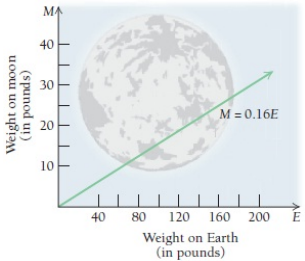
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A linear function is given by

$$y = mx + b \quad \text{or} \quad f(x) = mx + b$$

and has a graph that is the straight line parallel to the graph of $y = mx$ *and crossing the* $y-$*axis at* $(0, b)$*. The point* $(0, b)$ *is called the* $y-$**intercept**

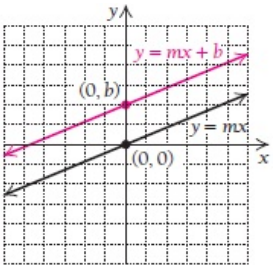
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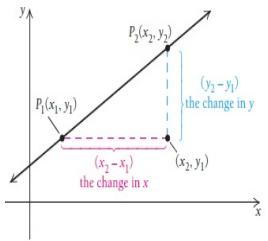
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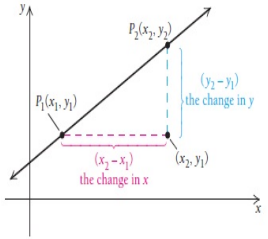


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Theorem

The slope of a line containing points (x_1, y_1) *and* (x_2, y_2) *is* $m = \dfrac{y_2 - y_1}{x_2 - x_1} = \dfrac{\textit{change in } y}{\textit{change in } x}$.

Applications of slope

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Applications of slope

Slope has many real-world applications. For example, numbers like 2%, 3%, and 6% are often used to represent the grade of a road, a measure of how steep a road on a hill is. A 3% grade (3%=3/100) means that for every horizontal distance of 100 ft, the road rises 3 ft.

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Applications of linear functions: Example Business (Profit-and-Loss Analysis)

When a business sells an item, it receives the *price* paid by the consumer (this is normally greater than the *cost* to the business of producing the item).

a) The **Total revenue** that a business receives is the product of the number of items sold and the price paid per item. Thus, if Raggs, Ltd., sells x suits at \$80 per suit, the total revenue $R(x)$, in dollars, is given by

$$R(x) = (\text{Unit price}) \times (\text{Quantity sold}) = 80x.$$

If $C(x) = 20x + 10000$, graph R and C using the same set of axes.

b) The **total profit** that business receives is the amount left after all costs have been subtracted from the total revenue. Thus, if $P(x)$ represents the total profit when x items are producted and sold, we have

$$P(x) = (\text{Total revenue}) - (\text{Total costs}) = R(x) - C(x).$$

Determine $P(x)$ and draw its graph using the same set of axes as was used for the graph in part a).

c) The company will *break even* at that value of x for which $P(x) = 0$ (that is, no profit and no loss). This is the point at which $R(x) = C(x)$. Find the break-even value of x.

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Outline

1. Functions
2. Graphs of functions
3. **Nonlinear Functions and models**
4. Mathematical modeling and curve fitting

Definition

A quadratic function f is given by

$$f(x) = ax^2 + bx + c, \quad \text{where } a \neq 0.$$

The graph of a quadratic function $f(x) = ax^2 + bx + c$ is called a **parabola**.

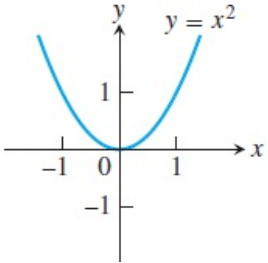
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Nonlinear Functions and models

Definition

A function p is a polynomial if

$$p(x) = a_n x^n + a_{n-1} x^{n-1} + ... + a_1 x + a_0$$

where n is a nonnegative integer and the numbers $a_0, a_1, ..., a_n$ are real constants (called the **coefficients** *of the polynomial).*

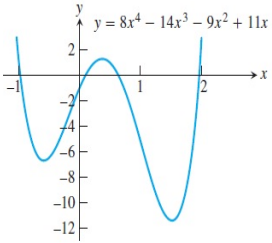
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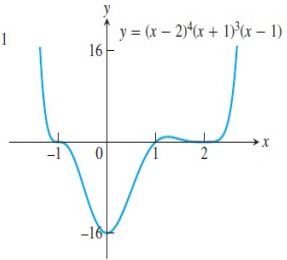
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A rational function is a quotient or ratio of two polynomials:

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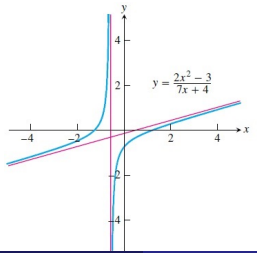
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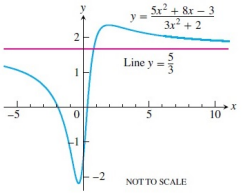
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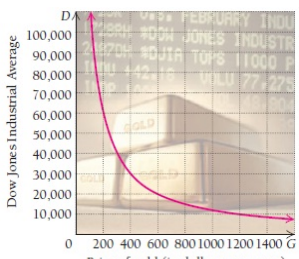
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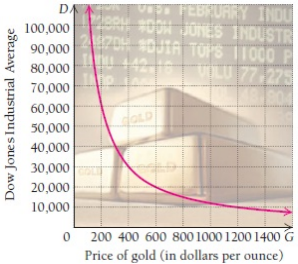


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Absolute-Value Functions $y = |x|$, Square-Root Functions $y = \sqrt{x}$.

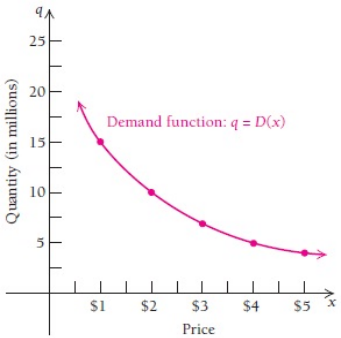
Supply and Demand Functions

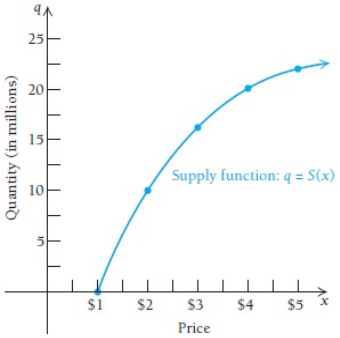
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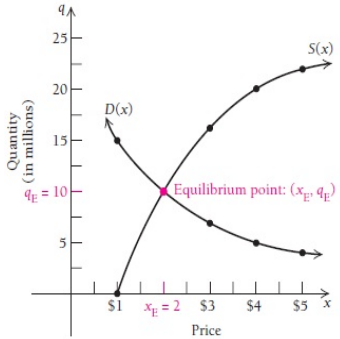
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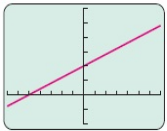
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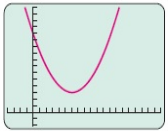
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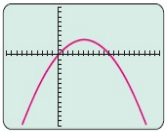
Linear function:
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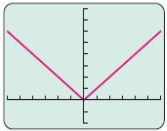
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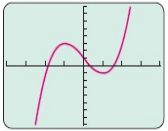
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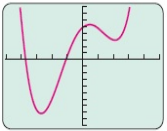
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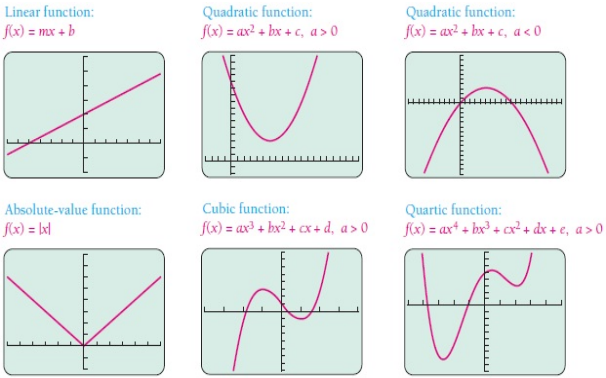


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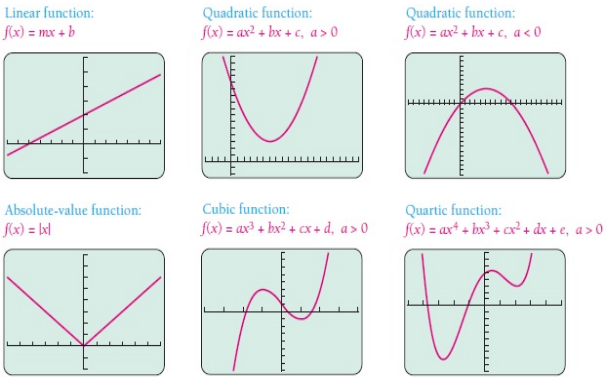
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Fitting Functions to Data



Let's consider some real-world data. How can we decide which, if any, type of function might fit the data?

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One simple way is to examine a graph of the data called a **scatterplot**.

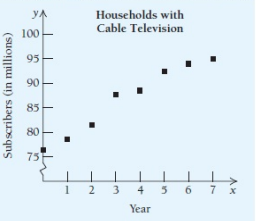
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Mathematical modeling and curve fitting

Example Business: Cable TV Subscribers. The following table shows the number of U.S. households with cable television and a scatterplot of the data. It appears that the data can be represented or modeled by a linear function.

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Year, x	Number of Cable Television Subscribers (in millions), y	Scatterplot
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2001, 2	81.5	
2002, 3	87.8	
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(*Source*: Nielsen Media Research.)

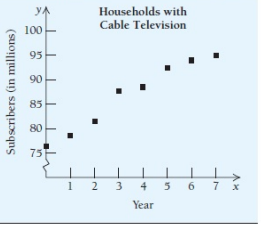
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Scatterplot



(*Source*: Nielsen Media Research.)

a) Find a linear function that fits the data.

b) Use the model the predict the number of cable TV subscribers in 2014.

Mathematical modeling and curve fitting

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